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1 Object Diagram - Admin Context

UML Type: Object Diagram **Category:** Admin **Source:** docs/diagrams/objects/admin/admin_objects.puml

1.1 Overview

This object diagram captures a runtime snapshot of the admin domain, showing how an Admin user interacts with models, reports, audit logs, and stats.

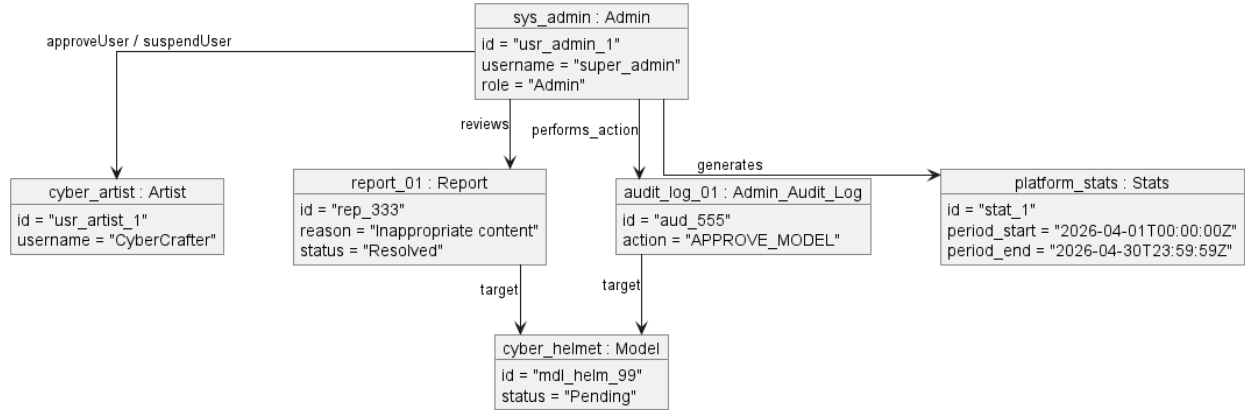


Figure 1: Object Diagram - Admin Context

1.2 Objects

1.2.1 sys_admin : Admin

The platform administrator instance.

Attribute	Value
id	"usr_admin_1"
username	"super_admin"
role	"Admin"

1.2.2 cyber_artist : Artist

An artist whose account and models are subject to admin actions.

Attribute	Value
id	"usr_artist_1"
username	"CyberCrafter"

1.2.3 cyber_helmet : Model

A model currently awaiting admin review.

Attribute	Value
id	"mdl_helm_99"
status	"Pending" (Model_Status.PENDING)

1.2.4 report_01 : Report

A resolved abuse report targeting the model.

Attribute	Value
id	“rep_333”
reason	“Inappropriate content”
status	“Resolved” (Report_Status.REVIEWED)

1.2.5 audit_log_01 : Admin_Audit_Log

An audit log entry created when the admin acted on the model.

Attribute	Value
id	“aud_555”
action	“APPROVE_MODEL” (Audit_Action.APPROVE_MODEL)

1.2.6 platform_stats : Stats

A periodic platform statistics snapshot generated by the admin.

Attribute	Value
id	“stat_1”
period_start	“2026-04-01T00:00:00Z”
period_end	“2026-04-30T23:59:59Z”

1.3 Relationships

From	Relationship	To	Description
sys_admin	performs_action	audit_log_01	Admin’s action is recorded in the audit log
audit_log_01	target	cyber_helmet	The model is the target of the audit action
sys_admin	reviews	report_01	Admin processes the pending report
report_01	target	cyber_helmet	The report points to the model
sys_admin	generates	platform_stats	Admin triggers stats aggregation
sys_admin	approveUser / suspendUser	cyber_artist	Admin manages artist account status

1.4 System Behavior Notes

- When an admin approves a model, the system calls `model.update({ status: APPROVED })` and creates an `Admin_Audit_Log` entry. The diagram shows both the approval action (`audit_log action = APPROVE_MODEL`) and the model's earlier pending state.
- The `Report` record with `status = "Resolved"` reflects a `Report` that the admin has reviewed and closed. This maps to `Report_Status.REVIEWED` in the schema.
- The `platform_stats` object captures a full month period. The `data` field (not shown here) would contain a JSON structure with totals for orders, revenue, new users, and top models.
- Admin audit logs are append-only. The `admin_id` FK on `Admin_Audit_Log` connects back to the admin user, enabling the audit log viewer in the dashboard.

2 Object Diagram - Auth Context

UML Type: Object Diagram Category: Auth Source: docs/diagrams/objects/auth/auth_objects.puml

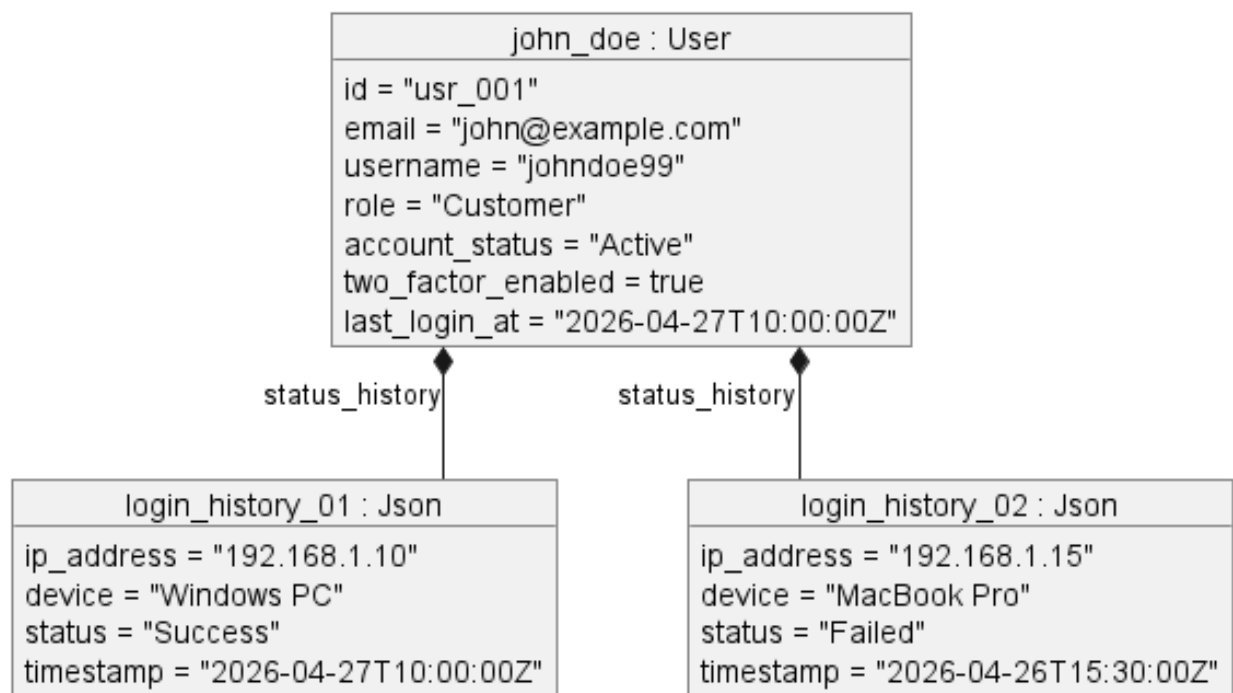


Figure 2: Object Diagram - Auth Context

2.1 Overview

This object diagram illustrates the runtime state of a `User` and their associated `status_history` JSON entries after two login attempts. It models the auth-context snapshot as it exists in the database.

2.2 Objects

2.2.1 john_doe : User

A concrete instance of the `User` entity representing an authenticated customer.

Attribute	Value
id	"usr_001"
email	"john@example.com"
username	"johndoe99"
role	"Customer"
account_status	"Active" (APPROVED in schema)
two_factor_enabled	true
last_login_at	"2026-04-27T10:00:00Z"

Note: `account_status = "Active"` in the diagram corresponds to `Account_Status.APPROVED` in the Prisma schema.

2.2.2 login_history_01 : Json

A JSON entry embedded in the `status_history` array on the User record. Represents a successful login event.

Attribute	Value
ip_address	"192.168.1.10"
device	"Windows PC"
status	"Success"
timestamp	"2026-04-27T10:00:00Z"

2.2.3 login_history_02 : Json

A second embedded JSON entry representing a failed login attempt.

Attribute	Value
ip_address	"192.168.1.15"
device	"MacBook Pro"
status	"Failed"
timestamp	"2026-04-26T15:30:00Z"

2.3 Relationships

From	Relationship	To	Description
john_doe (User)	status_history (composition)	login_history_01 (Json)	First entry in the status_history Json array
john_doe (User)	status_history (composition)	login_history_02 (Json)	Second entry in the status_history Json array

The composition notation indicates that `login_history_01` and `login_history_02` are owned by and cannot exist independently from `john_doe`. In practice, they are stored directly as JSON objects in the `status_history` column.

2.4 System Behavior Notes

- The `status_history` field is a `Json[]` column on the `User` table in Prisma. When an admin approves or rejects a user, a new JSON record is appended to this array.
- This diagram represents the snapshot after one successful session and one failed attempt, showing that the platform tracks session events in-place on the User record.
- The `last_login_at` field is updated by the `AuthService` on every successful login, as shown in the `sequence_local_login.puml` diagram.

3 Object Diagram - Catalog Context

UML Type: Object Diagram **Category:** Catalog **Source:** docs/diagrams/objects/catalog/catalog_objects.puml

3.1 Overview

This object diagram represents the runtime state of the catalog domain. It shows a specific 3D model instance linked to its creator (Artist), its category, and its tags.

3.2 Objects

3.2.1 `cyber_artist` : Artist

The creator of the featured model.

Attribute	Value
id	"usr_artist_1"
username	"CyberCrafter"
role	"Artist"

3.2.2 `cat_scifi` : Category

The category the model belongs to.

Attribute	Value
id	"cat_001"
name	"Sci-Fi"
slug	"sci-fi"

3.2.3 `tag_cyberpunk` : Tag

A descriptive tag applied to the model.

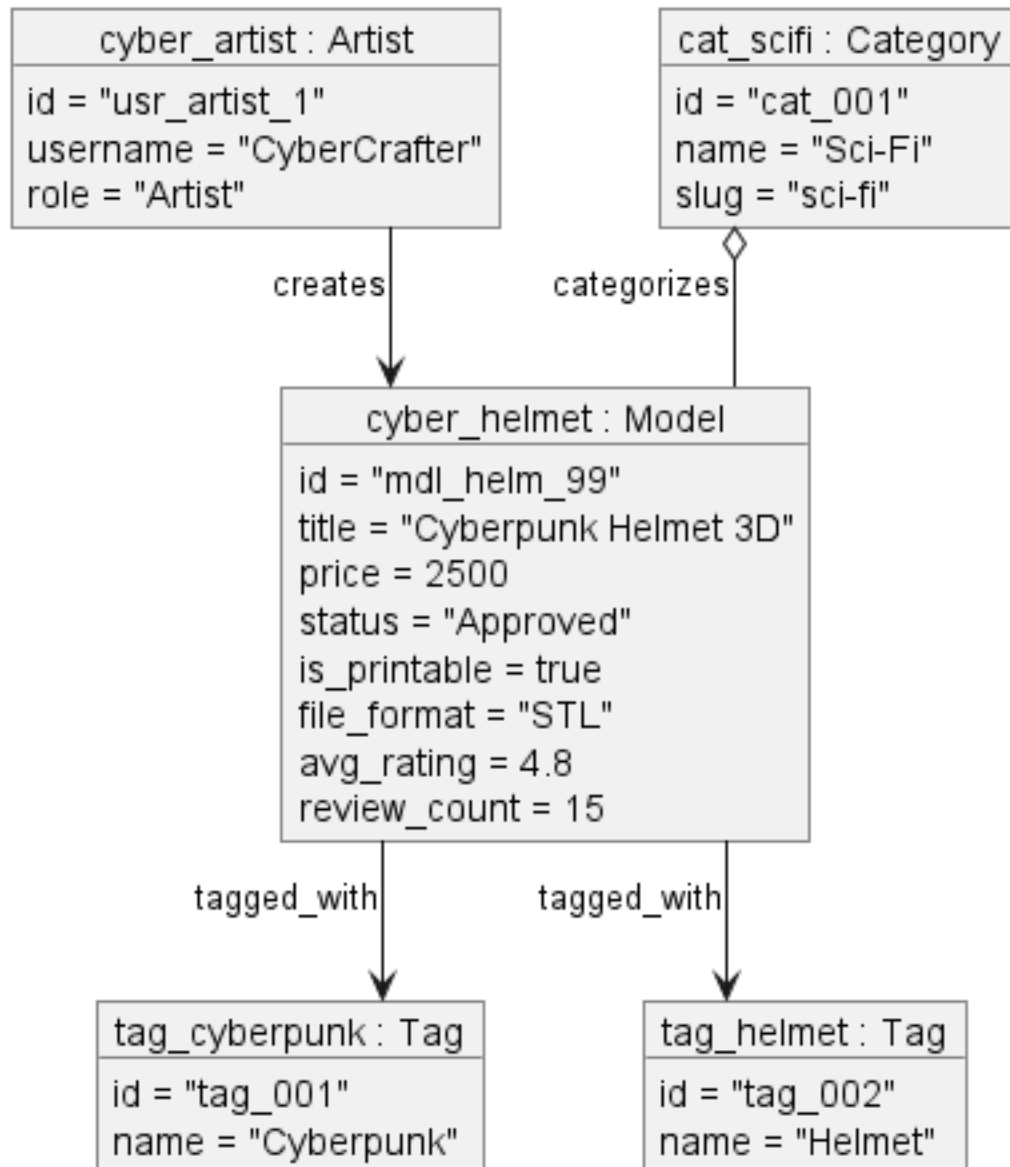


Figure 3: Object Diagram - Catalog Context

Attribute	Value
id	“tag_001”
name	“Cyberpunk”

3.2.4 tag_helmet : Tag

A second descriptive tag applied to the model.

Attribute	Value
id	“tag_002”
name	“Helmet”

3.2.5 cyber_helmet : Model

The 3D model at the centre of this snapshot.

Attribute	Value
id	“mdl_helm_99”
title	“Cyberpunk Helmet 3D”
price	2500 (IDR)
status	“Approved” (Model_Status.APPROVED)
is_printable	true
file_format	“STL”
avg_rating	4.8
review_count	15

Note: The file_format field shows “STL” in this object diagram. The Prisma schema defaults to “glb” for uploaded models. This is technically inconsistent - the system accepts GLB/GLTF files via the upload flow. STL would need explicit handling.

3.3 Relationships

From	Relationship	To
cyber_artist	creates	cyber_helmet
cat_scifi	categorizes	cyber_helmet
cyber_helmet	tagged_with	tag_cyberpunk
cyber_helmet	tagged_with	tag_helmet

The **categorizes** link is represented with an aggregation (open diamond) notation indicating that a Category can exist without any models. The **creates** link is a directed association from Artist to Model. Tags are connected via a many-to-many implicit join table managed by Prisma.

3.4 System Behavior Notes

- The `status = "Approved"` state means this model is publicly visible in the catalog to non-admin users.
- The `avg_rating` and `review_count` are denormalized fields updated after each review submission by aggregating the `Review` table.
- The `is_printable = true` flag makes this model available for selection in the Print Order flow.

4 Object Diagram - Commerce Context

UML Type: Object Diagram **Category:** Commerce **Source:** docs/diagrams/objects/commerce/commerce_object

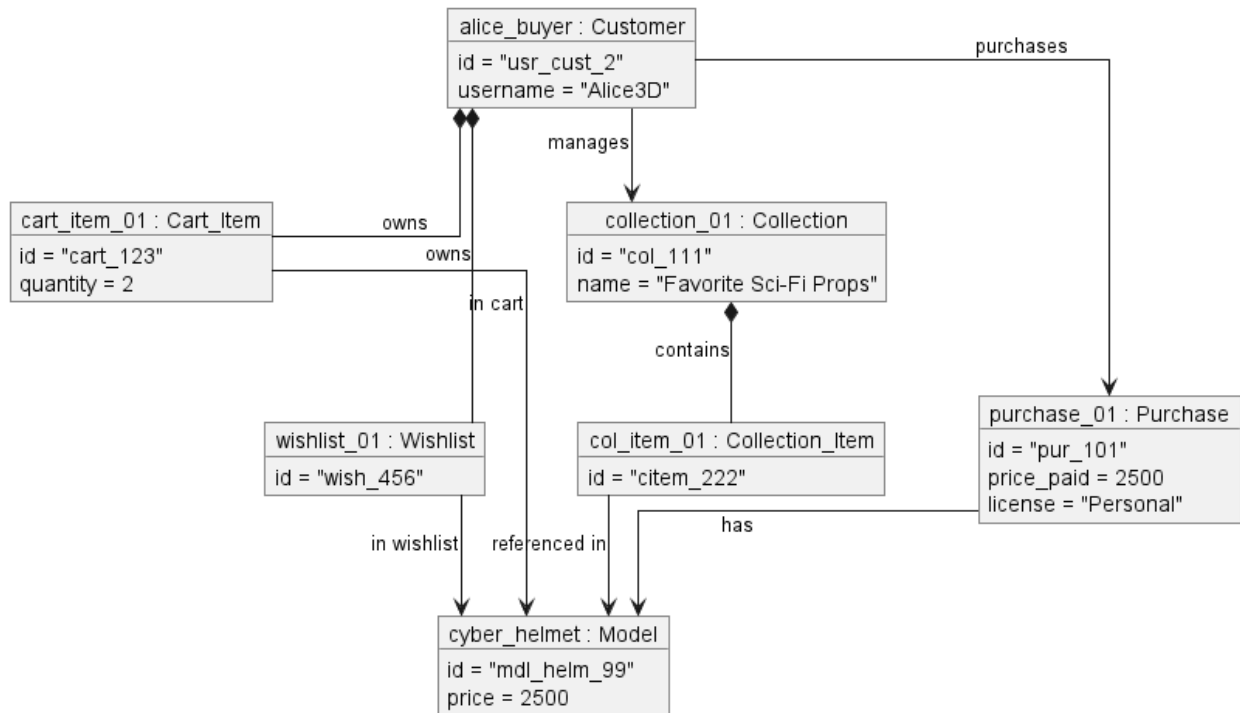


Figure 4: Object Diagram - Commerce Context

4.1 Overview

This object diagram shows a runtime snapshot of the commerce domain. It models a customer's cart, wishlist, collection, and completed purchase for a single model.

4.2 Objects

4.2.1 alice_buyer : Customer

The purchasing user.

Attribute	Value
id	“usr_cust_2”
username	“Alice3D”

4.2.2 cyber_helmet : Model

The 3D model in question.

Attribute	Value
id	“mdl_helm_99”
price	2500 (IDR)

4.2.3 cart_item_01 : Cart_Item

A cart entry for the model.

Attribute	Value
id	“cart_123”
quantity	2

Note: The diagram shows quantity = 2. In the real system, the cart normalization logic forces quantity = 1 for all digital models (`cart_Item.updateMany` sets quantity to 1 on every cart fetch). This object is a conceptual representation.

4.2.4 wishlist_01 : Wishlist

A wishlist entry for the model.

Attribute	Value
id	“wish_456”

4.2.5 collection_01 : Collection

A named collection owned by the customer.

Attribute	Value
id	“col_111”
name	“Favorite Sci-Fi Props”

4.2.6 col_item_01 : Collection_Item

The join record linking the collection to the model.

Attribute	Value
id	"citem_222"

4.2.7 purchase_01 : Purchase

The confirmed purchase record created after payment success.

Attribute	Value
id	"pur_101"
price_paid	2500 (IDR)
license	"Personal" (License_Type.PERSONAL_USE)

4.3 Relationships

From	Relationship	To	Description
alice_buyer	owns (composition)	cart_item_01	Cart item belongs to customer
cart_item_01	in cart	cyber_helmet	Model referenced in cart
alice_buyer	owns (composition)	wishlist_01	Wishlist entry belongs to customer
wishlist_01	in wishlist	cyber_helmet	Model saved to wishlist
alice_buyer	manages	collection_01	Customer owns the collection
collection_01	contains (composition)	col_item_01	Collection item belongs to collection
col_item_01	referenced in	cyber_helmet	Model referenced in collection
alice_buyer	purchases	purchase_01	Confirmed ownership record
purchase_01	has	cyber_helmet	Purchase links to the model

4.4 System Behavior Notes

- In the real system, a user can simultaneously have a model in their wishlist, cart, collection, and purchase records. These are four independent tables in Prisma.
- The **Purchase** record is only created by the Midtrans webhook handler after **transaction_status = "settlement"**. A cart item alone does not mean the user owns the model.
- The **Collection_Item** is deleted via **CASCADE** if either the parent **Collection** or the referenced **Model** is deleted.
- The **Wishlist** entry is similarly **CASCADE**-deleted if the model is deleted.

5 Object Diagram - Complete System Snapshot

UML Type: Object Diagram **Category:** Uncategorized (Complete System Overview) **Source:** docs/diagrams/objects/complete_system_object_diagram.puml

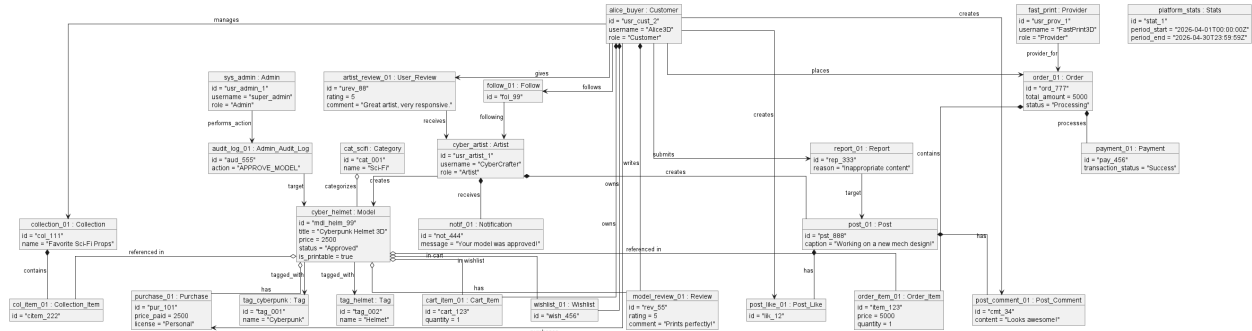


Figure 5: Object Diagram - Complete System Snapshot

5.1 Overview

This is the master object diagram for the entire 3Dex platform. It instantiates one representative object from every major domain and connects them to illustrate how all parts of the system interact in a single coherent runtime snapshot.

5.2 Objects by Domain

5.2.1 Users

Object Name	Type	Key Attributes
sys_admin	Admin	id="usr_admin_1", username="super_admin", role="Admin"
cyber_artist	Artist	id="usr_artist_1", username="CyberCrafter", role="Artist"
alice_buyer	Customer	id="usr_cust_2", username="Alice3D", role="Customer"
fast_print	Provider	id="usr_prov_1", username="FastPrint3D", role="Provider"

5.2.2 Catalog and Models

Object Name	Type	Key Attributes
cat_scifi	Category	id="cat_001", name="Sci-Fi"
tag_cyberpunk	Tag	id="tag_001", name="Cyberpunk"

Object Name	Type	Key Attributes
tag_helmet	Tag	id="tag_002", name="Helmet"
cyber_helmet	Model	id="mdl_helm_99", title="Cyberpunk Helmet 3D", price=2500, status="Approved", is_printable=true

5.2.3 Shopping and Orders

Object Name	Type	Key Attributes
purchase_01	Purchase	id="pur_101", price_paid=2500, license="Personal"
cart_item_01	Cart_Item	id="cart_123", quantity=1
wishlist_01	Wishlist	id="wish_456"
order_01	Order	id="ord_777", total_amount=5000, status="Processing"
order_item_01	Order_Item	id="item_123", price=5000, quantity=1
payment_01	Payment	id="pay_456", transaction_status="Success"

5.2.4 Social and Community

Object Name	Type	Key Attributes
model_review_01	Review	id="rev_55", rating=5, comment="Prints perfectly!"
artist_review_01	User_Review	id="urev_88", rating=5, comment="Great artist, very responsive."
post_01	Post	id="pst_888", caption="Working on a new mech design!"
post_like_01	Post_Like	id="lik_12"
post_comment_01	Post_Comment	id="cmt_34", content="Looks awesome!"
follow_01	Follow	id="fol_99"
collection_01	Collection	id="col_111", name="Favorite Sci-Fi Props"
col_item_01	Collection_Item	id="citem_222"

5.2.5 System and Admin

Object Name	Type	Key Attributes
report_01	Report	id="rep_333", reason="Inappropriate content"
notif_01	Notification	id="not_444", message="Your model was approved!"

Object Name	Type	Key Attributes
audit_log_01	Admin_Audit_Log	id="aud_555", action="APPROVE_MODEL"
platform_stats	Stats	id="stat_1", period_start="2026-04-01", period_end="2026-04-30"

5.3 Relationships

5.3.1 Catalog and Model Relationships

From	Relationship	To
cyber_artist	creates	cyber_helmet
cat_scifi	categorizes	cyber_helmet
cyber_helmet	tagged_with	tag_cyberpunk
cyber_helmet	tagged_with	tag_helmet

5.3.2 Commerce Relationships

From	Relationship	To
alice_buyer	purchases	purchase_01
purchase_01	has	cyber_helmet
alice_buyer	owns (composition)	cart_item_01
cart_item_01	in cart	cyber_helmet
alice_buyer	owns (composition)	wishlist_01
wishlist_01	in wishlist	cyber_helmet
alice_buyer	places	order_01
fast_print	provider_for	order_01
order_01	contains (composition)	order_item_01
order_item_01	referenced in	cyber_helmet
order_01	processes (composition)	payment_01

5.3.3 Social Relationships

From	Relationship	To
alice_buyer	writes (composition)	model_review_01
model_review_01	has	cyber_helmet
alice_buyer	gives	artist_review_01
artist_review_01	receives	cyber_artist
cyber_artist	creates (composition)	post_01
alice_buyer	creates	post_like_01
alice_buyer	creates	post_comment_01
post_01	has (composition)	post_like_01
post_01	has (composition)	post_comment_01
alice_buyer	follows	follow_01
follow_01	following	cyber_artist
alice_buyer	manages	collection_01

From	Relationship	To
collection_01	contains (composition)	col_item_01
col_item_01	referenced in	cyber_helmet

5.3.4 System and Admin Relationships

From	Relationship	To
alice_buyer	submits	report_01
report_01	target	post_01
cyber_artist	receives (composition)	notif_01
sys_admin	performs_action	audit_log_01
audit_log_01	target	cyber_helmet

5.4 System Behavior Notes

- This diagram is the union of all sub-category object diagrams. It demonstrates the full graph of dependencies in a single user journey where Alice browses, purchases, and reviews a model, while the artist posts content, the provider fulfills a print job, and the admin oversees everything.
- All `id` values are representative slugs, not real UUIDs, for readability.
- The `status = "Processing"` on `order_01` is a conceptual simplification. In the actual schema, this maps to intermediate states tracked by `Order_Item.print_status` rather than the `Order.status` field (which uses PENDING/PAID/FAILED/CANCELLED).

6 Object Diagram - Notifications Context

UML Type: Object Diagram **Category:** Notifications **Source:** docs/diagrams/objects/notifications/notifications

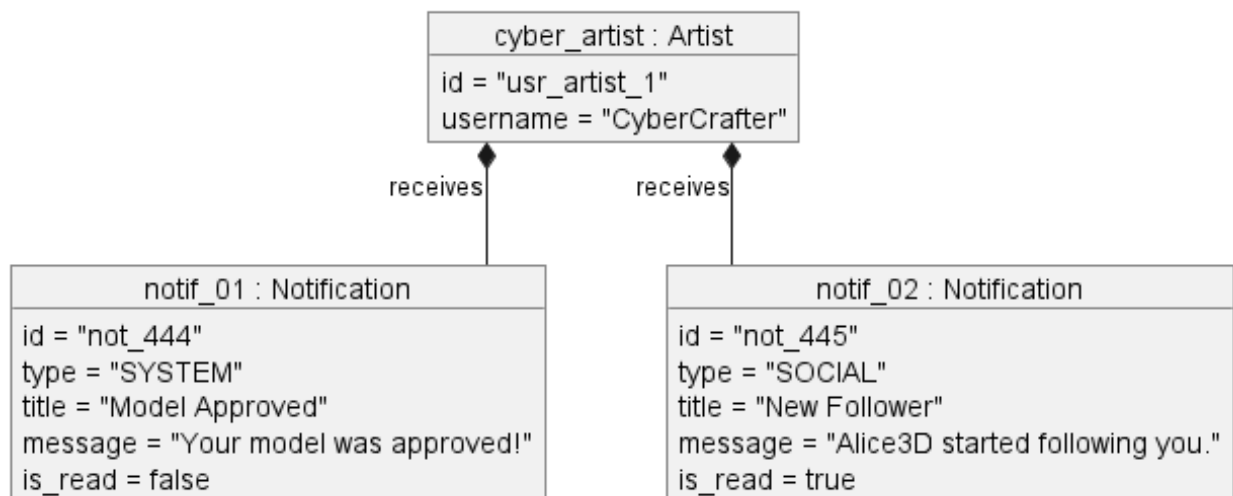


Figure 6: Object Diagram - Notifications Context

6.1 Overview

This object diagram shows a snapshot of the notifications domain, modeling two distinct notification records delivered to a single Artist user.

6.2 Objects

6.2.1 cyber_artist : Artist

The notification recipient.

Attribute	Value
id	"usr_artist_1"
username	"CyberCrafter"

6.2.2 notif_01 : Notification

A system notification triggered by an admin action.

Attribute	Value
id	"not_444"
type	"SYSTEM"
title	"Model Approved"
message	"Your model was approved!"
is_read	false

6.2.3 notif_02 : Notification

A social notification triggered by a follow action.

Attribute	Value
id	"not_445"
type	"SOCIAL"
title	"New Follower"
message	"Alice3D started following you."
is_read	true

6.3 Relationships

From	Relationship	To	Description
cyber_artist	receives (composition)	notif_01	System notification owned by artist
cyber_artist	receives (composition)	notif_02	Social notification owned by artist

6.4 System Behavior Notes

- The **type** field is a free-form string in the schema. The values “SYSTEM” and “SOCIAL” are application-level conventions used by the `NotificationService`.
- The **is_read** = **false** on **notif_01** means it will appear highlighted in the notification panel and contribute to the unread badge count.
- Notifications are created by `NotificationService.create_notification()` and dispatched from multiple points in the application: the payment webhook, the follow handler, and the print job lifecycle manager.
- The **data** field (not shown here) can carry optional JSON metadata, for example a model ID or post ID that the notification relates to, enabling deep-link navigation.

7 Object Diagram - Print Context

UML Type: Object Diagram **Category:** Print **Source:** docs/diagrams/objects/print/print_objects.puml

7.1 Overview

This object diagram captures a runtime snapshot of the print job domain, modeling a full print order lifecycle from customer submission through payment. It shows the relationships between the customer, provider, order, items, configuration, payment, and model.

7.2 Objects

7.2.1 `alice_buyer` : Customer

The customer who submitted the print order.

Attribute	Value
id	“usr_cust_2”
username	“Alice3D”

7.2.2 `fast_print` : Provider

The approved 3D printing service provider assigned to fulfill the job.

Attribute	Value
id	“usr_prov_1”
username	“FastPrint3D”
role	“Provider”

7.2.3 `order_01` : Order

The print job order record.

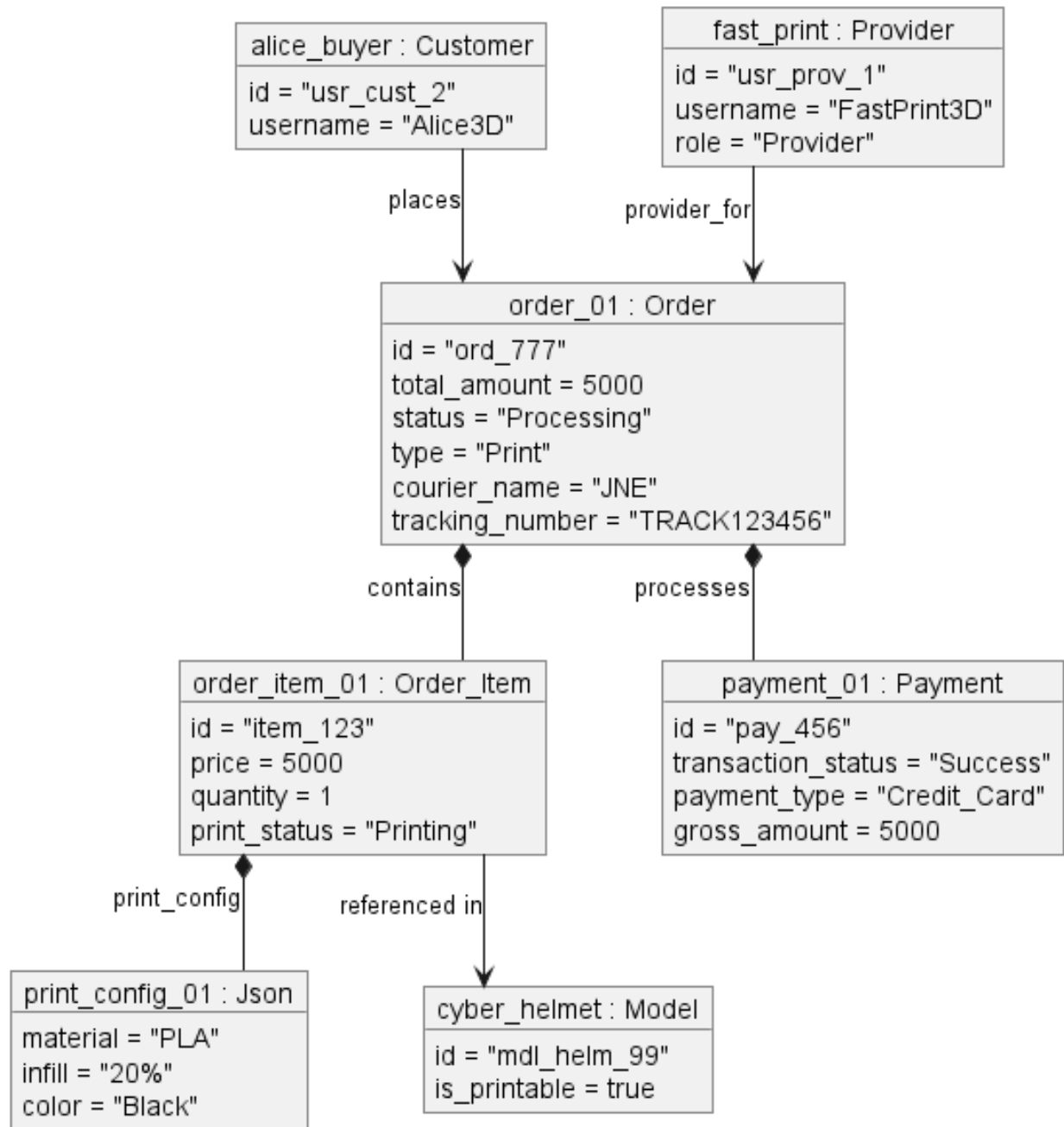


Figure 7: Object Diagram - Print Context

Attribute	Value
id	"ord_777"
total_amount	5000 (IDR)
status	"Processing" (Order_Status.PENDING or custom state)
type	"Print" (Order_Type.PRINT_JOB)
courier_name	"JNE"
tracking_number	"TRACK123456"

Note: The diagram shows `status = "Processing"`. The schema does not have a "Processing" state in `Order_Status`. The actual valid values are PENDING, PAID, FAILED, and CANCELLED. Processing state is tracked at the `Order_Item` level via the `print_status` field (Print_Status enum: PENDING, ACCEPTED, PROCESSING, SHIPPED, DELIVERED, CANCELLED).

7.2.4 order_item_01 : Order_Item

A single line item in the print order.

Attribute	Value
id	"item_123"
price	5000 (IDR)
quantity	1
print_status	"Printing" (Print_Status.PROCESSING)

7.2.5 print_config_01 : Json

The print configuration JSON embedded in the `order_item_01` record.

Attribute	Value
material	"PLA"
infill	"20%"
color	"Black"

7.2.6 payment_01 : Payment

The successful Midtrans payment record for this order.

Attribute	Value
id	"pay_456"
transaction_status	"Success" (Midtrans "settlement" value)
payment_type	"Credit_Card"
gross_amount	5000 (IDR)

7.2.7 cyber_helmet : Model

The model being printed.

Attribute	Value
id	“mdl_helm_99”
is_printable	true

7.3 Relationships

From	Relationship	To	Description
alice_buyer	places	order_01	Customer creates the order
fast_print	provider_for	order_01	Provider is assigned to the order
order_01	contains (composition)	order_item_01	Item is part of the order
order_item_01	print_config (composition)	print_config_01	Config JSON embedded in item
order_item_01	referenced in	cyber_helmet	Item points to the model being printed
order_01	processes (composition)	payment_01	Payment record linked to the order

7.4 System Behavior Notes

- Print orders use `Order_Type.PRINT_JOB`. The `provider_id` foreign key on `Order` is set at creation time when the customer selects a provider.
- Print configuration (material, infill, color, scale) is stored in the `print_config` JSON column on each `Order_Item`. The provider reads this to understand what to produce.
- The `print_status` field on `Order_Item` tracks the physical lifecycle of each item independently (PENDING -> ACCEPTED -> SHIPPED -> DELIVERED or CANCELLED).
- The payment for a print job is separate from digital asset payments. Print jobs may use a different billing arrangement since the `payment_01` shown here has a nonzero `gross_amount`.

8 Object Diagram - Social Context

UML Type: Object Diagram **Category:** Social **Source:** docs/diagrams/objects/social/social_objects.puml

8.1 Overview

This object diagram models the social domain at runtime. It shows how an Artist and a Customer interact through posts, likes, comments, follows, and reviews.

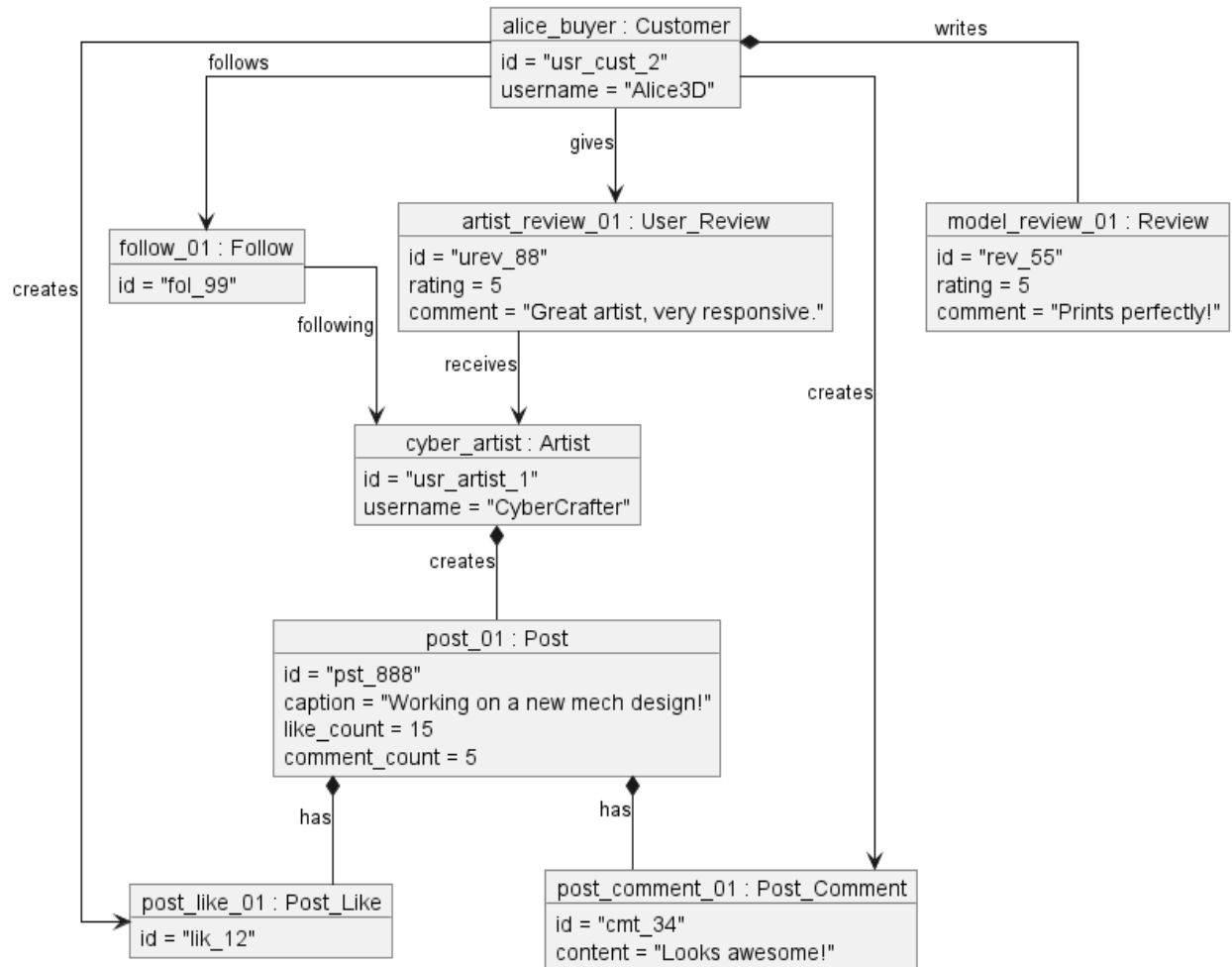


Figure 8: Object Diagram - Social Context

8.2 Objects

8.2.1 cyber_artist : Artist

The creator being followed and reviewed.

Attribute	Value
id	“usr_artist_1”
username	“CyberCrafter”

8.2.2 alice_buyer : Customer

The user interacting with the artist’s content.

Attribute	Value
id	“usr_cust_2”
username	“Alice3D”

8.2.3 post_01 : Post

A community post published by the artist.

Attribute	Value
id	“pst_888”
caption	“Working on a new mech design!”
like_count	15
comment_count	5

8.2.4 post_like_01 : Post_Like

A like record created by Alice on the post.

Attribute	Value
id	“lik_12”

8.2.5 post_comment_01 : Post_Comment

A comment left by Alice on the post.

Attribute	Value
id	“cmt_34”
content	“Looks awesome!”

8.2.6 follow_01 : Follow

A directed follow edge from Alice to the artist.

Attribute	Value
id	“fol_99”

8.2.7 model_review_01 : Review

A model review written by Alice for one of the artist’s models.

Attribute	Value
id	“rev_55”
rating	5
comment	“Prints perfectly!”

8.2.8 artist_review_01 : User_Review

A user-level review of the artist as a seller, written by Alice.

Attribute	Value
id	“urev_88”
rating	5
comment	“Great artist, very responsive.”

8.3 Relationships

From	Relationship	To	Description
cyber_artist	creates (composition)	post_01	Artist owns the post
alice_buyer	creates	post_like_01	Alice created the like
alice_buyer	creates	post_comment_01	Alice created the comment
post_01	has (composition)	post_like_01	Like is part of the post
post_01	has (composition)	post_comment_01	Comment is part of the post
alice_buyer	follows	follow_01	Alice initiated the follow
follow_01	following	cyber_artist	The follow points to the artist
alice_buyer	writes (composition)	model_review_01	Review owned by Alice
alice_buyer	gives	artist_review_01	User review given by Alice
artist_review_01	receives	cyber_artist	Artist receives the user review

8.4 System Behavior Notes

- `like_count` and `comment_count` on `Post` are denormalized counters. They are incremented/decremented atomically in the database alongside the create/delete of `Post_Like` and `Post_Comment` records.
- `Review` (model review) and `User_Review` (person review) are two separate tables. A `Review` requires a prior `Purchase`, while a `User_Review` requires either a purchase from the artist or a completed print job from the provider.
- The `Follow` table is directional: `follower_id` is Alice, `following_id` is the artist. When Alice follows the artist, a `Notification` is created for the artist.
- Post creation is restricted to users with `role = ARTIST` or `role = PROVIDER`. Customers cannot create posts.